Exploring Optimization Strategies for Strongly Convex Ridge Regression under Noise and Conditioning

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Inspiration

- Linear regression is among the most widely used supervised machine learning techniques

Given a dataset with feature matrix $X \epsilon \mathbb{R}^{n \times d}$ and response vector $y \epsilon \mathbb{R}^{n}$

Goal: Estimate parameters $w \epsilon \mathbb{R}^{d}$ that minimize Mean Square Error (MSE)

MSE: $f(w) = min_{w} \frac{1}{n} ||Xw - y||^{2}$

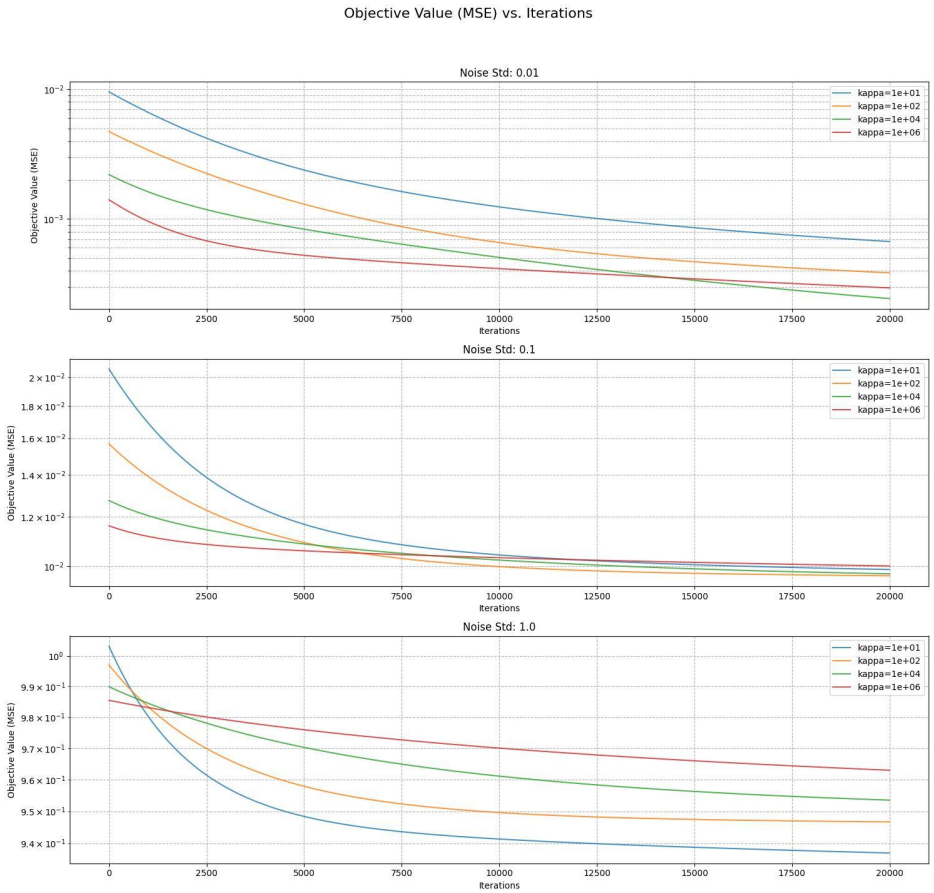
- The MSE objective is convex, but
  - Performance depends on the feature matrix X, especially its conditioning
  - Multicollinearity (highly correlated features) makes $X^T X$ ill-conditioned
  - Ill-conditioning leads to slow convergence for optimization methods like gradient descent

- Real-world datasets are often ill-conditioned and noisy, motivating analysis of how different gradient descent methods perform under these conditions

- We will test Gradient Descent, Ridge Gradient Descent, and Ridge Stochastic Gradient Descent on datasets with varying conditioning and noise levels

Standard Gradient Descent

- Standard Gradient Descent shows monotonic decrease in the MSE objective across all experimental settings
- The condition number of the design matrix strongly influences convergence speed
  - well-conditioned problems converge rapidly
  - ill-conditioned problems require significantly more iterations
- As conditioning number increases, the objective curves flatten earlier
- Increasing noise levels raise the minimum achievable objective value, causing optimization to plateau sooner regardless of conditioning
- In high-noise settings, the objective quickly reaches a noise-dominated floor, where further iterations provide little to no improvement.
- The negative effects of ill-conditioning and noise together lead to both slower convergence and reduced attainable accuracy



Effect of Ridge Regularization (Ridge GD/SGD)

- The ridge regression objective introduces an L2 regularization term that improves conditioning and guarantees strong convexity:

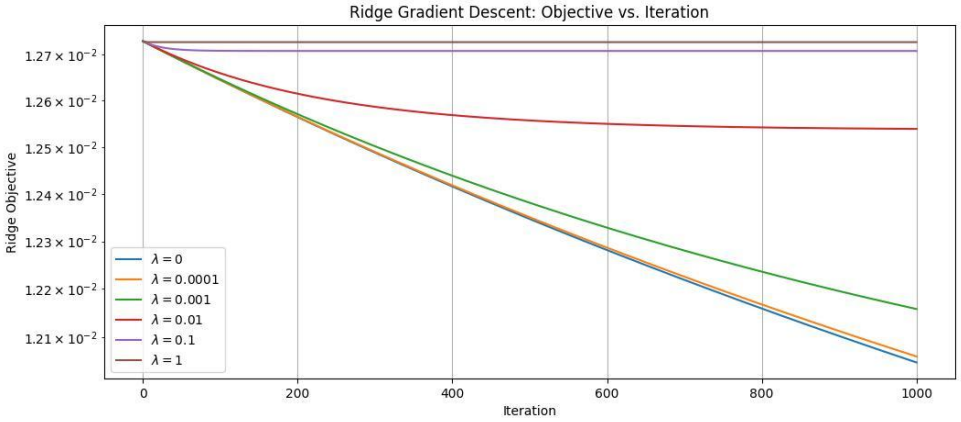
Objective: $f(w) = \frac{1}{n}||Xw - y||^2 + \lambda||w||^2, \lambda > 0$

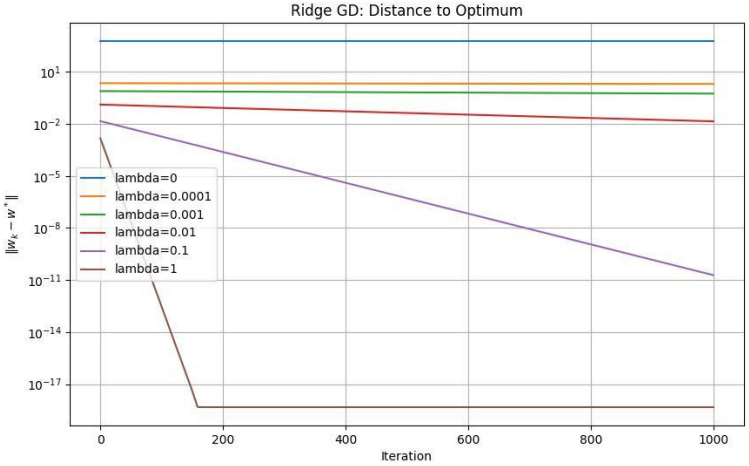
Hessian: $\nabla^2 f(w) = \frac{2}{n}X^T X + 2\lambda I$

- Introducing L2 regularization leads to the ridge objective, optimized via GD and SGD, with performance evaluated against the closed-form solution:

$$w^* = \left(\frac{X^T X}{n} + \lambda I\right)^{-1} \frac{X^T y}{n}$$

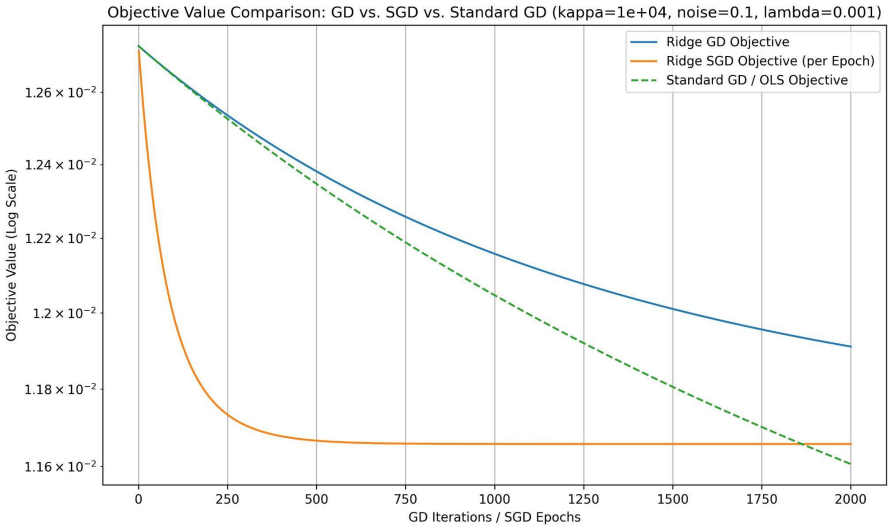
- The addition of the penalty term modifies the Hessian from $(1/n)X^TX$ to $(1/n)X^TX + \lambda I$, which improves the condition number by shifting eigenvalues, resulting in faster and more stable convergence

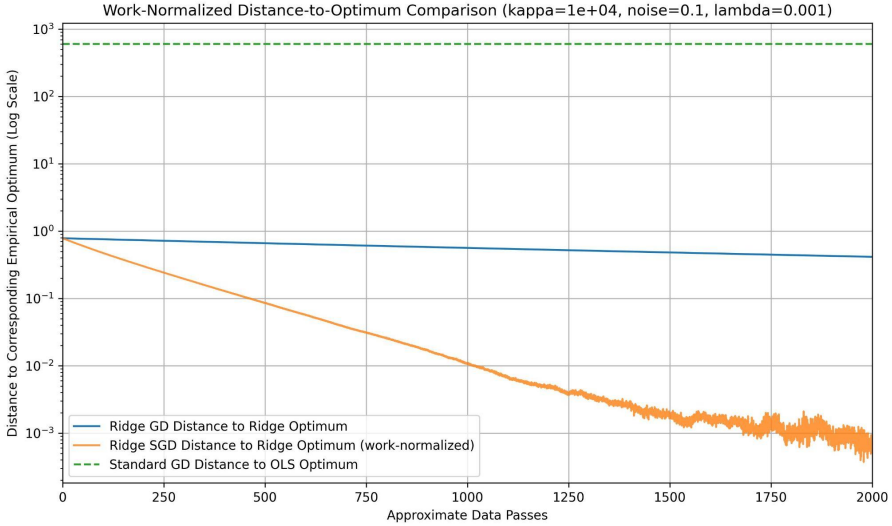




- As the regularization parameter λ increases, Gradient Descent exhibits faster convergence and earlier stabilization, but this introduces bias, leading to a higher final objective value compared to weakly regularized or unregularized cases

- Ridge regularization acts as an implicit preconditioner where larger λ accelerates the decay of the distance, while small λ retains ill-conditioning, causing slow progress along directions associated with small eigenvalues





- Ridge SGD achieves a fast initial drop in the objective, as stochastic gradients provide frequent updates that quickly reduce error, while ridge GD follows a smoother, deterministic trajectory and standard GD converges slowest due to the ill-conditioned objective
- The presence of regularization in ridge methods improves the Hessian leads to better conditioning and faster convergence compared to unregularized GD
- As iterations progress, ridge SGD begins to plateau due to gradient variance, while ridge GD continues a steady decrease toward the optimum, highlighting the tradeoff between speed and stability
- Using a work-normalized metric (effective data passes), ridge SGD still shows the fastest initial reduction in distance to optimum, but exhibits noisy asymptotic behavior, while ridge GD converges more consistently and standard GD again shows slow decay due to ill-conditioning

Theoretical and Empirical Convergence Rates

- For strongly convex quadratic objectives, theory predicts linear convergence given as:

$$||w_k - w^*|| \leq Cp^k$$

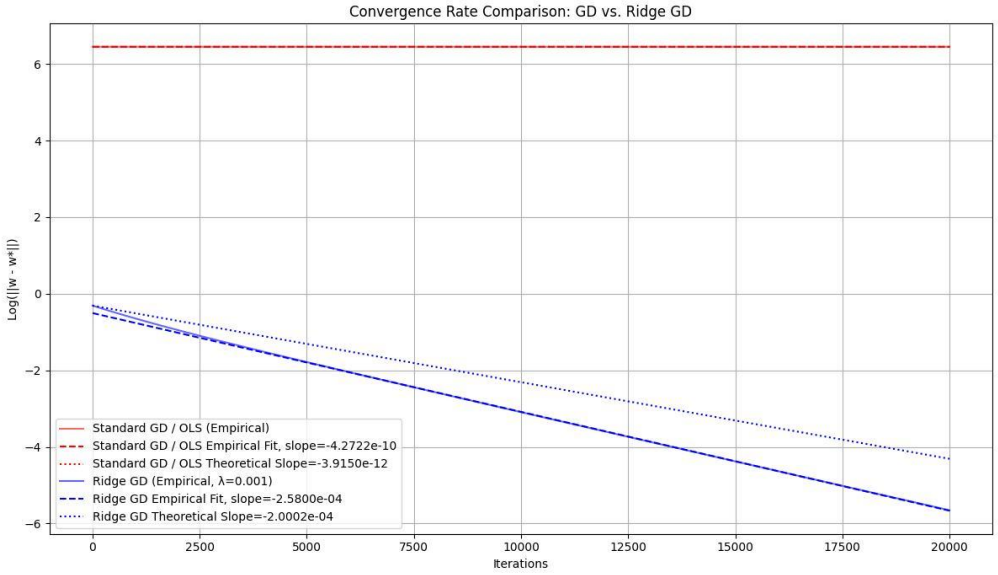
$$p \epsilon (0,1),\ p = max\{|1 - \eta\mu|, |1 - \eta L|\}$$

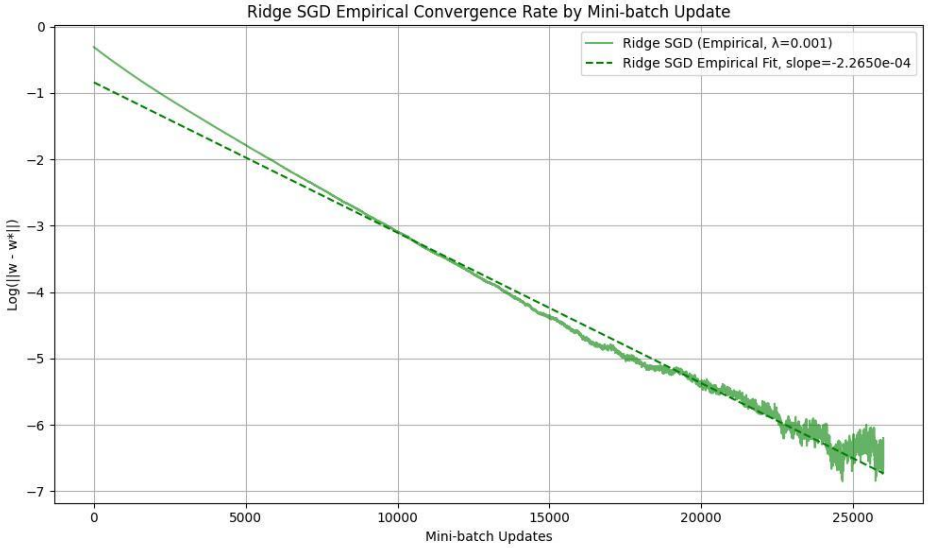
μ = smallest value of Hessian

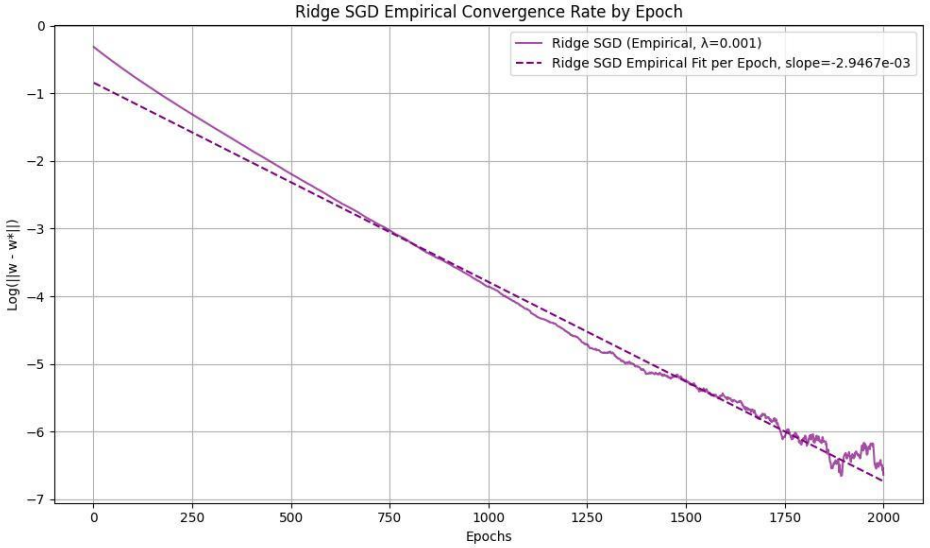
L = largest value of Hessian

- In the ridge-regularized setting, plotting log||w_k−w∗|| versus iterations reveals a clear linear trend, with slopes closely matching theory, showing that improved conditioning from λ directly accelerates convergence

- In contrast, the unregularized OLS problem shows extremely slow decay, consistent with its large condition number, confirming that poor spectral properties hinder convergence speed



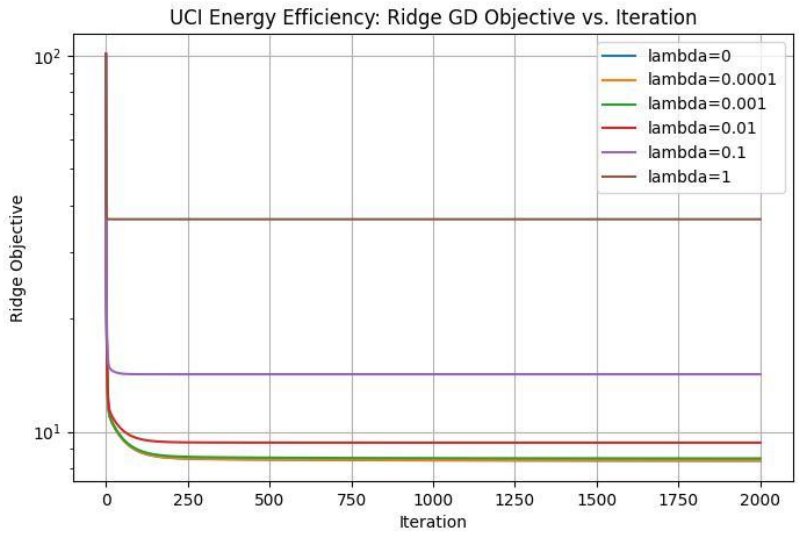


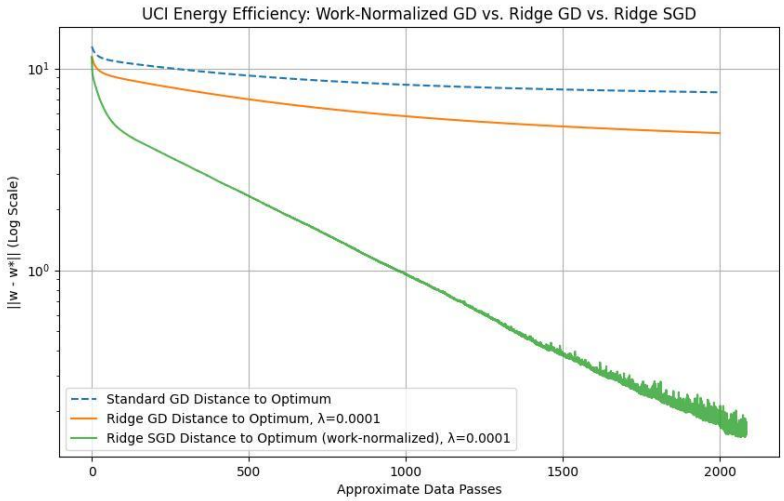


- For SGD, empirical convergence exhibits an initial linear decay, after which progress becomes slower and noisier due to stochastic effects
- Early progress aligns with theory, but later stages are dominated by stochastic gradient variance, especially near the optimum
- Aggregating SGD progress at the epoch level smooths out noise and highlights the same pattern, illustrating the tradeoff between fast initial progress and noisy behavior

Real-World Dataset Analysis (UCI Dataset)

- The UCI Energy Efficiency dataset consists of residential building simulations with eight input features (e.g., surface area, wall area, glazing), and targets heating and cooling loads
  - These features are naturally correlated, introducing induce ill-conditioning in the design matrix
- Applying Gradient Descent, ridge GD, and SGD to the UCI Energy Efficiency dataset allows us to test whether insights from synthetic experiments hold in a realistic setting with correlated features and uncontrolled conditioning
- As in the synthetic case, increasing the regularization parameter λ improves conditioning of the Hessian $(1/n)X^TX + \lambda I$, resulting in faster convergence and earlier stabilization
  - But this also introduces bias, leading to a higher final objective value





- For small λ, the optimization behavior closely resembles the unregularized case, with slower convergence driven by feature correlations and poor conditioning inherent in the dataset
- Ridge regularization consistently improves convergence over standard GD by modifying the spectral properties of the problem, while ridge SGD achieves the fastest initial reduction in distance to optimum under a work-normalized metric
- Near the optimum, ridge SGD shows higher variance and slower asymptotic convergence due to stochastic gradient noise, while ridge GD maintains smooth and stable convergence, reinforcing the tradeoff between speed and stability in practical settings

Key Findings

- The convergence rate of Gradient Descent is fundamentally governed by the conditioning of the objective, with ill-conditioned problems leading to slow progress
- Ridge regularization improves conditioning by modifying the Hessian $(1/n)X^TX + \lambda I$, resulting in significantly faster and more stable convergence
- There is a clear bias-variance tradeoff introduced by regularization
  - increasing λ accelerates convergence and stabilizes optimization, but raises the final objective value and limits the attainable accuracy due to added bias
- Stochastic Gradient Descent achieves the fastest initial progress in reducing both objective value and distance to the optimum, even under work-normalized comparisons, but suffers from noisy updates and slower asymptotic convergence, whereas ridge GD provides smoother, more consistent convergence across both synthetic and real-world datasets

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